RR-C-430B
May 19, 1971
SUPERSEDING
Fed. Spec. RR-C-430A
October 31, 1966

FEDERAL SPECIFICATION

CLIMBER SET, TREE AND POLE: STEEL, INTERCHANGEABLE GAFF, W/NYLON-NEOPRENE COATED STRAPS

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers an adjustable tree and pole climber set with interchangeable gaffs and nylon neoprene coated straps.

1.1.1 Federal specification coverage. Federal specifications do not include all types of commodities indicated by the titles of the specifications, but are intended to cover only those generally used by the Federal Government.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-636 - Box, Fiberboard.
PPP-B-640 - Boxes, Fiberb oard, Corrugated, Triple-Wall.

Federal Standard:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General

FSC 4240

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Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO., Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, and Standards from established distribution points in their agencies.)

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-130 - Identification Marking of U.S. Military Property.

(Copies of Military Specifications and Standards required by contractors in connection with specification procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply:

American Society for Testing and Materials (ASTM) Standards:

E18 - Test for Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials.
E138 - Wet Magnetic Particle Inspection.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

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Edison Electric Institute:

AP-2 - Specifications for Linemen's Climbing Equipment.

(Application for copies should be addressed to the Edison Electric Institute, 750 Third Avenue, New York, NY 10017.)

Society of Automotive Engineers (SAE):

SAE J404 - Chemical Compositions of SAE Alloy Steels.

(Application for copies should be addressed to the Society of Automotive Engineers, Two Pennsylvania Plaza, New York, NY 10001.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Tariff Publishing Officer, Room 202 Union Station, 516 W. Jackson Blvd., Chicago, IL 60606.)

National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the National Motor Freight Traffic Association, Inc., Agent, 1616 P Street, N.W., Washington, DC 20036.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal Agencies.)

3. REQUIREMENTS

3.1 Description. The climber set shall conform to figure 1 and EEI AP-2, except that the straps shall conform to figures 2 and 3, and the calf pad assembly shall conform to figure 4. The climber leg iron shall be adjustable from 14-3/4 to 19-1/4 inches in increments of 1/4 inch. The adjustable sleeve shall be held to the leg iron by two fasteners at any specified increment.

3.2 First article (preproduction model). When specified (see 6.2), the supplier shall furnish three climber sets and sample straps for eximation

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and test within the time frame specified, to prove prior to starting production, that his production methods will produce climber sets that comply with the requirements of this specification. Examination and tests shall be those specified in section 4 and shall be subject to surveillance and approval by the Government (see 6.3).

3.3 Materials. Materials shall be as specified in EEI AP-2, unless otherwise specified herein. Materials not specified shall be selected by the supplier and shall be subject to all provisions of this specification.

3.3.1 Leg irons. Leg irons shall be SAE 8630 or SAE 8640 steel forged with a quenched and tempered structure. Leg irons shall conform to required shapes of the finished pieces. The leg iron shall be heat treated after forging to R_c 34-38. The heat treatment shall consist of an austenitizing treatment, a quench sufficiently rapid to avoid non-martensitic products, and a tempor to reduce hardness to the required range. Steel properties of the finished leg iron shall be as follows:

Elongation in 2 inches	15 percent minimum
Hardness	34-38 Rockwell "C"
Yield strength	130,000 psi minimum

The stress of the leg iron shall not exceed seven-tenths of the yield strength when tested as specified in 4.5.2.5.

3.3.2 Strap material. The fabric used for the climber straps shall consist of four plies of plain weave, close-woven, nylon cloth of adequate thread count folded so that no raw edges are exposed. The cloth shall be impregnated or coated with neoprene and vulcanized so that the plies are not separable except by chemical means or by applying a tensile force larger than 20 pounds per inch of width. The 4-ply fabric used for the straps shall have a breaking strength of not less than 1,400 pounds for a section free from buckle holes. The fabric shall have a buckle-holding strength of not less than 300 pounds without evidence of failure and shall tear with the direction of the load application when tested to failure.

3.4 Gaffs. One set of pole gaffs and one set of tree gaffs shall be furnished with each climber set. The tree gaffs and the pole gaffs shall be interchangeable.

3.5 Magnetic particle requirements. The gaffs and leg shanks shall be subjected to the magnetic particle test specified in 4.5.2.1 and shall show no evidence of cracks or other subsurface or surface discontinuities.

3.6 Identification marking.

3.6.1 Military. The climber set shall be identified in accordance with MIL-STD-130, including date of manufacture and lot number.

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3.6.2 Civil agencies. The climber set shall be permanently and clearly marked with the manufacturer's name, or a trademark of such known character that the source of manufacture and lot may be readily identified.

3.7 Instruction sheet. An instruction sheet covering gaff replacement procedure and proper sharpening procedure shall be included with each climber set.

3.8 Workmanship. All parts and components of the climber set shall be free of forging laps, seams, visible cracks, and cross grinding marks. Cutting edges of the gaffs shall be sharp and ground smooth.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Material inspection. The climber set supplier is responsible for insuring that materials used are manufactured, examined, and tested to comply with referenced specifications and standards.

4.2 Classification of inspection. Inspection shall be classified as follows:

(a) Preproduction inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).
(c) Inspection of preparation for delivery (see 4.6).

4.3 Preproduction inspection.

4.3.1 Examination. The climber set shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Test. The climber set and sample straps shall be tested as specified in 4.5.2.1 through 4.5.2.6. Failure of any test shall be cause for rejection.

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4.4 Quality conformance inspection.

4.4.1 Lot size. A lot shall consist of all climber sets offered for acceptance at one time produced by one manufacturer during one calendar month (or 30 continuous days), at one plant, from the same materials, using the same manufacturing process and under essentially the same manufacturing conditions.

4.4.2 Examination. Each climber set shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.4.3 Nondestructive tests.

4.4.3.1 Wet magnetic particle. Each climber set shall be tested as specified in 4.5.2.1. Failure of the test shall be cause for rejection.

4.4.3.2 Hardness. Sampling shall be in accordance with MIL-STD-105, inspection level I. Testing shall be as specified in 4.5.2.6 with AQL of 1.0 percent defective.

4.4.4 Destructive tests.

4.4.4.1 Straps. Six sample straps (three for each test) without buckles and buckle holes shall be selected from each lot for testing in accordance with 4.5.2.2 and 4.5.2.3. Three sample straps with buckles and holes shall be selected from each lot for testing in accordance with 4.5.2.4. Failure of any test shall be cause for rejection of the lot.

4.4.4.2 Leg iron. One sample climber set shall be selected from each lot for testing in accordance with 4.5.2.5. Failure of this test shall be cause for rejection of the lot.

4.5 Inspection procedure.

4.5.1 Examination. Each climber set shall be examined for the following defects:

101. Tree and pole gaffs not interchangeable.
102. Dimensions not as specified.
103. Adjustable sleeve not held to leg iron by two fasteners at any specified increment.
104. Material not as specified.
105. Presence of forging laps or seams in the material.
106. Gaffs not free from cross grinding marks, or cutting edges not ground smooth.
107. Presence of visible cracks.

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108. Missing parts, components, or instruction sheet.
109. Incomplete or incorrect assembly.
110. Date of manufacture and lot number not included.

4.5.2 Tests.

4.5.2.1 Wet magnetic particle. Each gaff, leg shank, stirrup and metallic attachments shall be tested in accordance with ASTM E138. The direction of the magnetic field shall be both circular and longitudinal. The sequence of operation shall be continuous. All parts shall be de-magnetized after inspection. Evidence of cracks or other discontinuities shall constitute failure of this test.

4.5.2.2 Strap. Three specimens each of straps without buckle holes shall be tested for tensile breaking strength. Each specimen shall be secured in a testing machine and load shall be applied with a steady pull at a rate of 2 inches per minute until the strap breaks. Failure of any specimen to withstand a minimum force of 1,400 pounds shall constitute failure of this test.

4.5.2.3 Strap material. Three sample straps (see 4.4.1) without buckle holes shall be tested for ply separation or "peel" strength. Each sample shall be cut so that purchase may be made on any two adjacent plies and a tensile force applied in opposite directions so as to test the "peel" strength of the neoprene coating. Each sample shall be secured in a testing machine and a load shall be applied with a steady pull at a rate of 2 inches per minute until the plies are separated. Failure of any sample to withstand a minimum force of 20 pounds per inch of width shall constitute failure of this test.

4.5.2.4 Buckled strap. Three sample straps with buckle holes shall be tested by placing the strap in the buckle with the buckle tongue through the strap buckle hole. The strap and buckle shall be placed in the testing machine, and a load shall be applied with a steady pull at a rate of 2 inches per minute until failure occurs. A buckled strap holding strength of less than 300 pounds or evidence of tearing in any direction other than in the direction of the load application shall constitute failure of this test.

4.5.2.5 Climber. The individual climber shall be tested in a configuration so that the resulting coplanar and concurrent forces from a statically applied load at the center of the stirrup and parallel with a maintained offset of 1-3/4 inch to the climber shank shall result in the resultant forces passing through the penetration point of the gaff. The three areas of maximum bending (tensile or compression stresses, with two above and one below the gaff are to be monitored with electrical resistance type strain gages. The climber shall be loaded in excess of its yield strength (tension and compression).

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4.5.2.5.1 Procedure. One climber each shall be tested at the minimum, medium and maximum lengths of their operational size range. Stresses recorded at 300 pounds static loading shall not exceed seven-tenths of the climber yield strength either tensile or compressive obtained at the maximum stress location. If seven-tenths of the yield strength is exceeded this shall constitute failure of the test.

4.5.2.6 Hardness. Rockwell hardness shall be determined in accordance with ASTM E18. Failure of any part of the leg iron including the stirrup to meet the hardness range of R_c 34-38 shall constitute failure of this test. Failure of the gaff to meet the hardness range of R_c 45-55 or having a variation of more than three points in the Rockwell hardness reading taken from the tip of the gaff to a point 1-1/4-inches back from the tip shall constitute failure of this test.

4.6 Inspection of preparation for delivery.

4.6.1 Quality conformance inspection of pack.

4.6.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.6.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.6.1.3 Examination. Samples selected in accordance with 4.6.1.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

111.	Materials and containers not as specified for level A or B. Each incorrect material or container shall be considered one defect.
112.	Contents not cushioned as specified for level A or B.
113.	Box not waterproof sealed as specified for level A.
114.	Strapping not as specified for level A.
115.	Gross weight or size exceeds the limitations of the box specification for level B.
116.	Marking missing, illegible, incorrect, or incomplete.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A, B, or C as specified (see 6.2).

5.1.1 Level A. Each complete climber set with instruction sheet shall be packaged in a close-fitting box conforming to PPP-B-636, W6c, style optional.

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The contents shall be cushioned to prevent the sharp points from puncturing the box and to prevent free movement. The box shall be waterproof sealed with tape as specified for slotted style boxes in the appendix to the box specification.

5.1.2 Level B. Each complete climber set with instruction sheet shall be packaged as specified in 5.1.1 for level A except the box shall be type CF, class domestic, variety SW, grade 125 and waterproof sealing of the box shall not be required.

5.1.3 Level C. Each complete climber set with instruction sheet shall be packaged in a close-fitting fiberboard box adequate to afford protection against damage from the supplier to the initial destination. The supplier may use his standard commercial packaging when it fulfills these requirements.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A. The climber sets packaged as specified in 5.1, shall be packed together in close-fitting boxes conforming to PPP-B-601, overseas type, style optional. Box closure and strapping shall be in accordance with the appendix to the box specification. Strapping shall be class B.

5.2.2 Level B.

5.2.2.1 Military agencies. Unless otherwise specified (see 6.2 and 6.4), level B packing shall be as follows: The climber sets packaged as specified in 5.1, shall be packed together in close-fitting boxes conforming to PPP-B-636, V3c, style optional or PPP-B-640, class 2, style optional. The gross weight or size of each box shall not exceed the limitations of the applicable box specification. Box closure and strapping shall be in accordance with the appendix to the applicable box specification.

5.2.2.2 Civil agencies. Twelve climbers packaged as specified in 5.1, shall be packed in a fiberboard box conforming to PPP-B-636, class domestic, style RSC.

5.2.3 Level C. The climber sets packaged as specified in 5.1, shall be packed in a manner to assure carrier acceptance and safe delivery to destination at lowest rating in a fiberboard container complying with Uniform Freight Classification rules or with National Motor Freight Classification rules.

5.3 Marking.

5.3.1 Civil agencies. In addition to any special marking required in the contract or order, interior packages and exterior shipping containers shall be marked in accordance with Fed. Std. No. 123.

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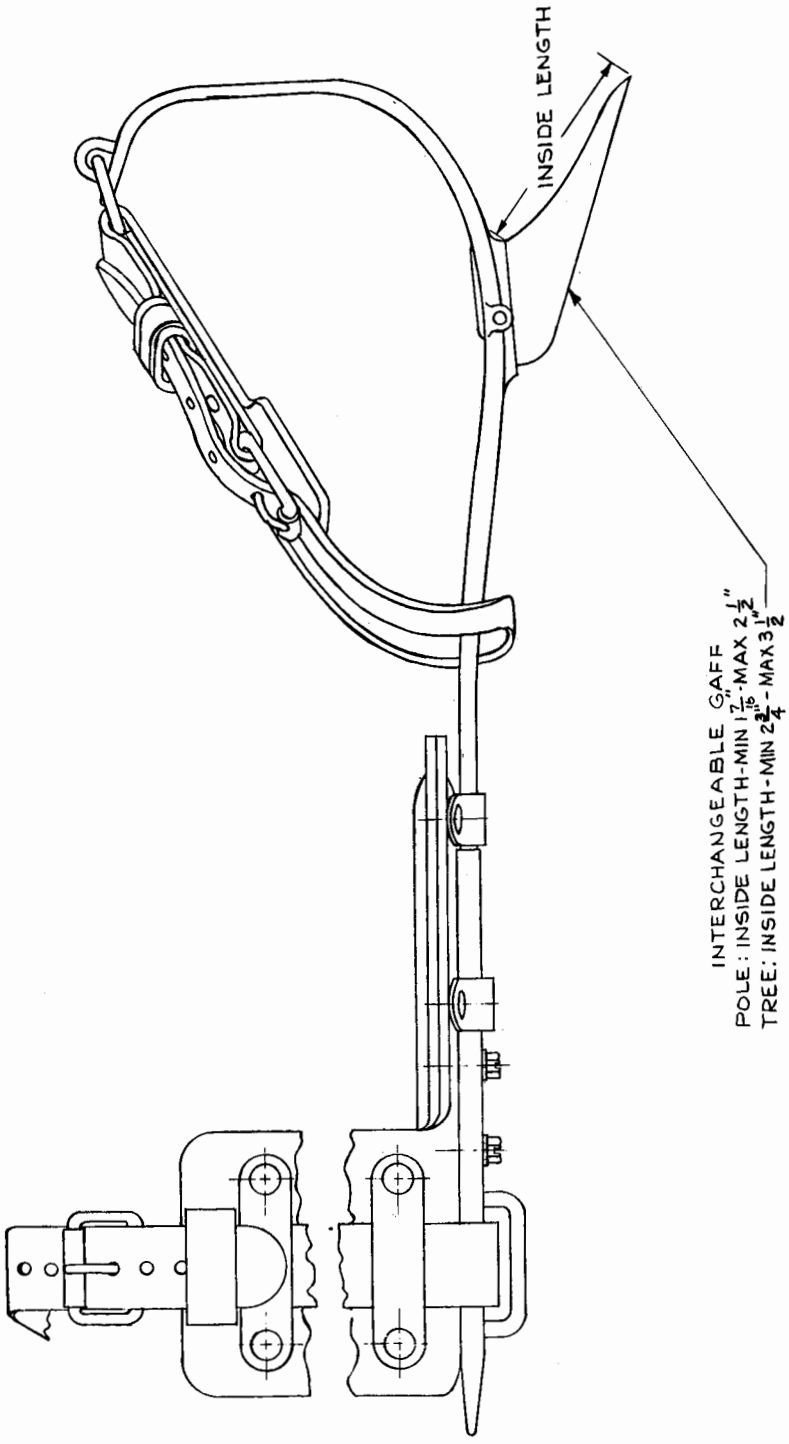


FIGURE 1 - CLIMBER SET, TREE AND POLE: STEEL INTERCHANGEABLE GAFF, WITH NYLON NEOPRENE COATED STRAPS, TYPICAL

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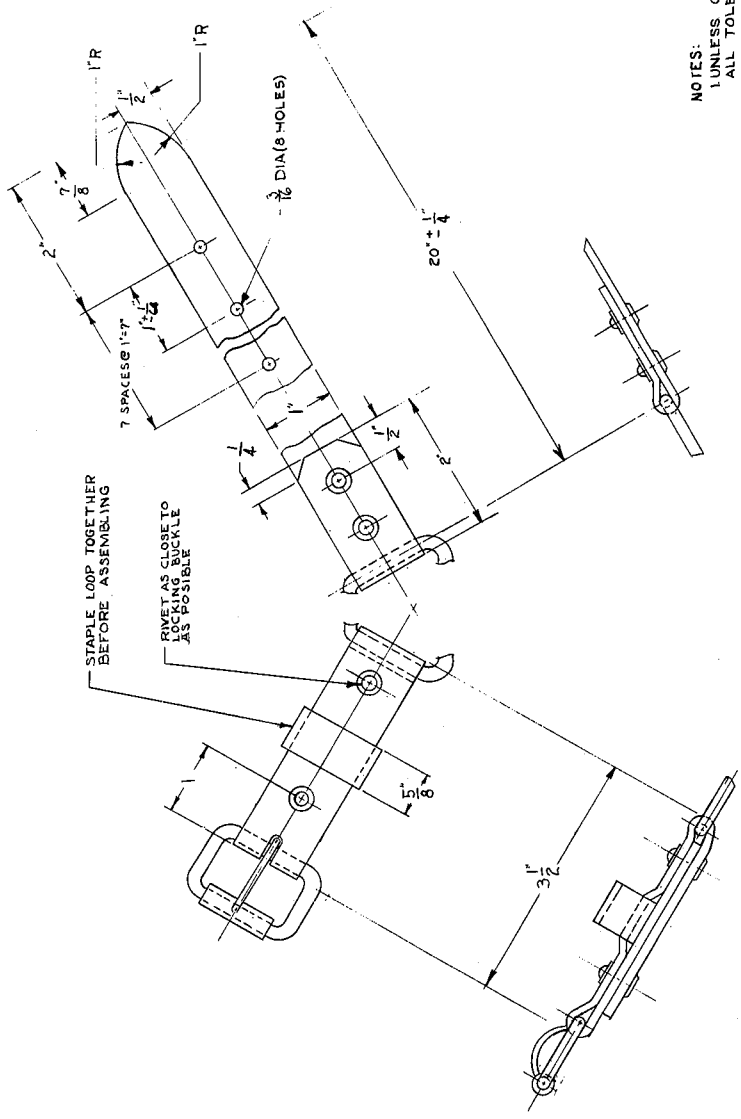
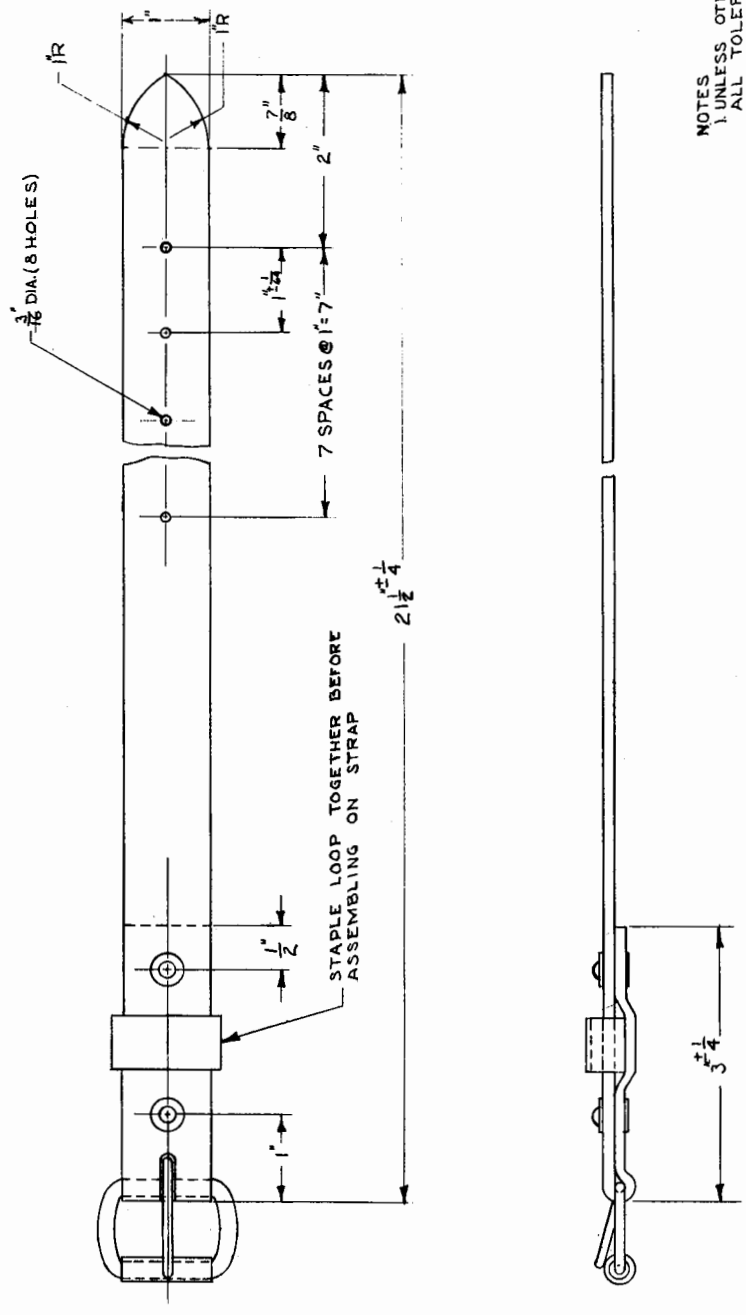


FIGURE 2 - ANKLE STRAP ASSEMBLY

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NOTES
1. UNLESS OTHERWISE SPECIFIED ALL TOLERANCES ARE ± 1/2".
2. STRAP MATERIAL: NYLON, NEOPRENE COATED.

FIGURE 3 - CALF STRAP ASSEMBLY

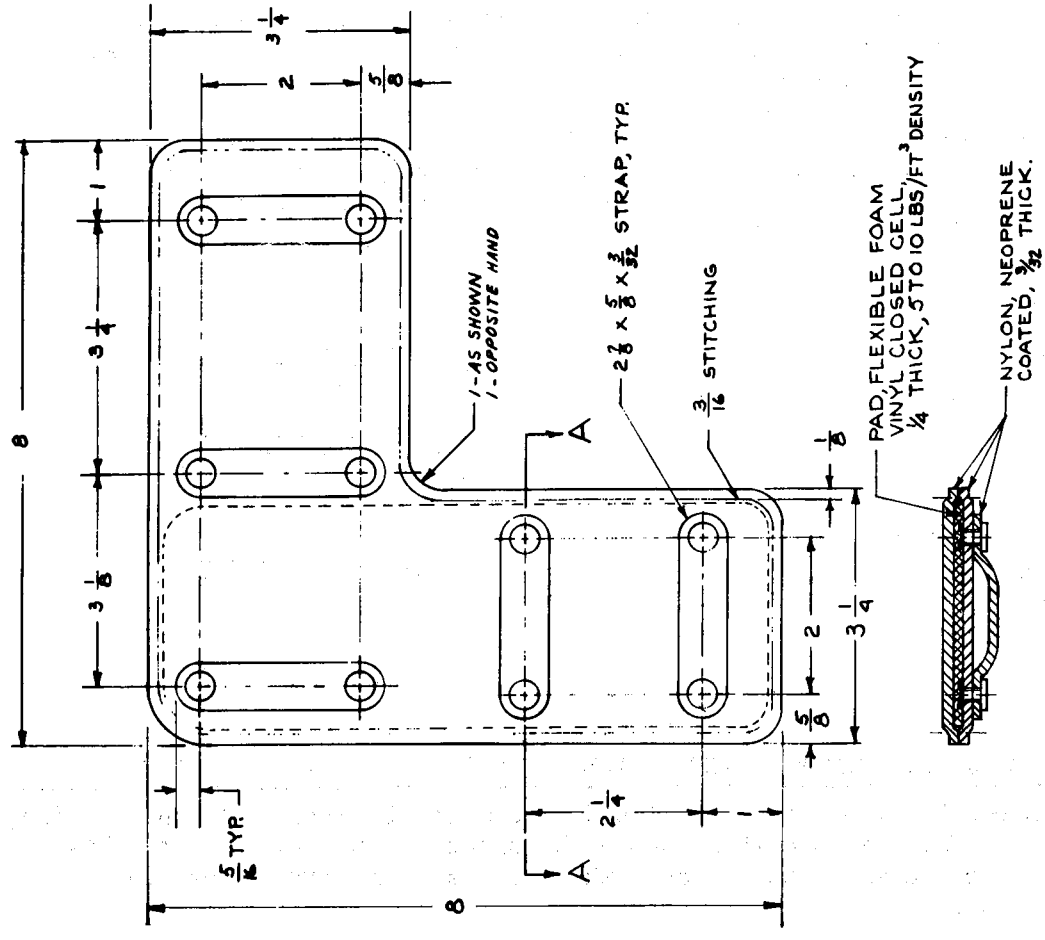
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NOTES:
1. UNLESS OTHERWISE SPECIFIED, ALL TOLERANCES ARE ± 1/32
2. MATERIAL: AS NOTED

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SECTION A-A

FIGURE 4 CALF PAD ASSEMBLY - TYPICAL

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5.3.2 Military agencies. In addition to any special marking required in the contract or order, interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. Climber sets are intended for use in ascending and descending trees and poles.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in the procurement documents:

(a) Title, number, and date of this specification.
(b) Time frame required for submission of first article when required (see 3.2).
(c) Level of packaging and level of packing required (see 5.1 and 5.2). Level B packaging and packing is intended to provide economical but limited protection and should be specified only when it is determined the climber sets will be held in covered storage.
(d) Special marking required (see 5.3.1 and 5.3.2).

6.3 Preproduction model. Any changes or deviations of production climber sets from the approved preproduction model during production will be subject to the approval of the contracting officer. Approval of the preproduction model will not relieve the supplier of his obligation to furnish climber sets conforming to this specification.

6.4 Level B packaging (see 5.1.2) has been developed by the Army and Air Force for conditions specified in 6.2 For Navy procurements, details of level B packaging, when required, should be developed at the time of procurement when all conditions are known.

MILITARY INTEREST:

Custodians:
Army - ME
Navy - YD
Air Force - 84

Review activities:
Army - SM, MU
Navy - MC

User activity:
Army - EL

Preparing activity:
Army - ME

CIVIL AGENCY INTEREST:
GSA - FSS

Project No. 4240-0308

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